1 Gaussian with unknown mean and variance

The likelihood function is an exponential family of distributions with a complete sufficient statistic $T = (T_1, T_2)$, where

$$T_1 = \sum_{i=1}^{n} X_i \text{ and } T_2 = \sum_{i=1}^{n} X_i^2$$

Thus by the Lehmann-Scheffe theorem, any unbiased estimators that are a function of $T = (T_1, T_2)$ are UMVUE for the unknown parameters μ and σ^2.

Therefore, the empirical mean

$$\hat{\mu} = \frac{1}{n} T_1$$

is an unbiased estimator of μ and hence is an UMVUE of μ. Recall that the same empirical-mean estimator is also an UMVUE of μ when the variance σ^2 is known, and the varaince of the estimator is given by

$$\mathrm{Var}_\theta[\hat{\mu}] = \frac{\sigma^2}{n}$$

which decreases linearly with the sample size n. To find an UMVUE for σ^2, recall that when the mean parameter μ is the known, an UMVUE is given by the empirical variance.

$$x$$

if μ is unknown, it is very tempting to replace μ by its UMVUE $\hat{\mu}$ and consider the estimator

$$\hat{\sigma^2}' = \frac{1}{n} \sum_{i=1}^{n} (X_i - \hat{\mu})^2$$

where

$$\hat{\mu} = \frac{1}{n} \sum j = 1^n X_j$$

Let $X_i = \sigma Z_i + \mu$, where $(Z_1, Z_2, \ldots, Z_n)$ are i.i.d. standard Gaussian varaibles. We can write

$$\begin{aligned} \hat{\sigma^2}' &= \frac{1}{n} \sum_{i=1}^{n} \left(X_i - \frac{1}{n} \sum_{j=1}^{n} X_j \right)^2 \\ &= \frac{\sigma^2}{n} \sum_{i=1}^{n} \left(Z_i - \frac{1}{n} \sum_{j=1}^{n} Z_j \right)^2 \end{aligned}$$

now $\mathbb{E}_\theta[\hat{\sigma^2}'] = \frac{\sigma^2}{n}(n-1) = \frac{n-1}{n}\sigma^2$. This estimator is biased for σ^2, but we can get rid of this by multiplying $\hat{\sigma^2}'$ with the scaling factor $(n-1)/n$.

Note that $\sum_{i=1}^{n} (X_i - \hat{\mu})^2 = T_2 - 2(\frac{T_1}{n})T_1 + n(\frac{T_1}{n})^2$ which is a function of $T = (T_1, T_2)$.

The variance of $\hat{\sigma^2}''$ is calculated by

$$\mathrm{Var}_\theta[\hat{\sigma^2}''] = ...$$

This is the price to pay for not knowing the value of the mean parameter μ.

When $n = 1$, X itself is a complete sufficient statistic. Therefore, if an unbiased estimator exists, it is also an UMVUE. As an exercise, show that for $n = 1$, an unbiased estimator for σ^2 does not exist (when both μ and σ^2 are unknown).

Need to find universal delta function such that $\mathbb{E}_\theta[\delta(x)] = \sigma^2 \ \forall(\mu, \sigma^2) = \int_{-\infty}^{\infty} \delta(x)\delta_\theta(x)\ dx = \sigma^2$.

1.1 Uniform with unknown support $X_i \sim \mathcal{U}([0,\theta])$

Recall that in this case,

$$T = \max_{i\in[n]} X_i$$

is a complete statistic for θ. Start by checking $\mathbb{E}_\theta[T] = ?$. $T \leq \theta$.

Recall that the pdf of T can be calculated as

$$f_\theta(t) = nt^{n-1}\theta^{-n}1_{\{0\leq t\leq\theta\}}$$

The expectation of T is thus given by

$$\mathbb{E}_\theta[T] = \int_0^\theta t f_\theta(t)dt = \frac{n\theta}{n-1}$$

The variance

$$\begin{aligned}\mathrm{Var}_\theta[T] &= \mathbb{E}_\theta[T^2] - (\mathbb{E}_\theta[T])^2 \\ &= \frac{n\theta^2}{n+1} - \left(\frac{n\theta}{n+1}\right)^2\end{aligned}$$

From here we can also compute the varnace of ...

1.2 UMVUE without a complete sufficient statistic

Rao-Blackwellization can be used to improve the variance of an unbiased estimator. Applying Rao-Blackwellization with a complete sufficient statistic leads to an UMVUE. What if a complete sufficient statistic does NOT exist?

In the previous cases, the model parameter is $\Theta = (0,\infty)$. Here, we shall assume that $\Theta = (1,\infty)$, i.e., we have the additional knowledge that the unknown parameter $\theta > 1$. Note that with the additional knowledge $\theta > 1$, it does not make sense for the estimator to take any value smaller than 1. It is natural to consider a modified estimator

$$\hat{\theta}' = \begin{cases} 1 & , \quad \text{if } T \leq 1 \\ \frac{n+1}{n}T & , \quad \text{if } T > 1 \end{cases}$$

For any $\theta > 1$, the expected value of $\hat{\theta}'$ can be calculated as

$$\begin{aligned}\mathbb{E}_\theta[\hat{\theta}'] &= \int_0^1 f_\theta(t)\,dt + \int_1^0 \frac{n+1}{n} t f_\theta(t)\,dt \\ &= n\theta^{-n}\int_0^1 t^{n-1}\,dt + (n+1)\theta^{-n}\int_1^0 t^n\,dt \\ &= \theta^{-n} + \theta^{-n}(\theta^{n+1} + 1)\end{aligned}$$

Therefore, if T is again a complete sufficient statistic in this case, then $\hat{\theta}'$ is an UMVUE of θ, knowing that $\theta > 1$. Unfortunately, in this case, T is no longer complete, as it can be seen as follows:

Let $\phi(T)$ be a function of T such that $\mathbb{E}_\theta[\phi(T)] = 0,\ \forall \theta > 1$. As before, it can be shown that we must have $\phi(T) = 0,\ \forall t > 1$. However, for $t \le 1$, $\phi(t)$ does not have to be a zero function to satistfy $\mathbb{E}_\theta[\phi(T)] = 0,\ \forall \theta > 1$.

For example, if we let

$$\phi(t) = \begin{cases} t - \frac{n}{n+1}, & \text{if } t \le 1 \\ 0, & \text{if } t > 1 \end{cases}$$

Apparently, we have $\mathbb{P}_\theta[\phi(T) = 0] < 1$,

1.3 Variance reduction via unbiased estimators of 0

Mathematically, for a given likelihood family $f_\theta(x)$, a statistic $U = U(X)$ is called an unbiased estimator of 0 if $\mathbb{E}_\theta[U] = 0$ for all $\theta \in \Theta$.